Coursework 2: General Linear Model (GLM) and Permutation Testing

Giuliano Costa **SN: 24225233** **March 12th, 2025**

Note: Code in the Jupyter notebook is organized by 1 cell per 1 question, for easy referencing. For specific details I mention `code like this.`

Part 1: GLM Unifies Classical Hypothesis Testing

1. Two-sample t-test for testing the hypothesis that two groups have different means:

   (a) Using SciPy, we can create two normally distributed random variables with different means but the same variance. Then we can take 20 samples and compute the sample mean and σ.

   ```
   stats.norm.rvs(loc=1.5, scale=0.2, random_state=random_seed, size=20)
   stats.norm.rvs(loc=2.0, scale=0.2, random_state=random_seed+1, size=20)
   ```

   Note, it is important both to add +1 to the seed for correct sample generation (see Fig. 1). Our means and σ are within range for the two groups:

   ```
   Grp 1 Mean: 1.4850 Grp 1 std: 0.2677
   Grp 2 Mean: 2.0303 Grp 2 std: 0.1840
   ```

   (b) SciPy has a method to compute a T-test for two independent sets of samples: `stats.ttest_ind`
   It tests for the null hypothesis that both samples have identical means. It assumes the variances are equal by default. We can reject the null hypothesis that both samples have the same mean, given our p-value is 9.17×10^{-9} (lower than 1%) and our t-statistic is -7.317 (first group's mean is lower than the second).

   (c) Compute the t-statistic using the GLM model:

   i. We can express group membership using a design matrix (Equation 1). The dimension of the column space is equal to the rank of the matrix. Where the rank of the matrix is the number of linearly independent columns in X. In our case, with two linearly independent columns, dim(C(X))=2. (We know they are linearly independent by visual inspection).

   ii. To project Y onto our column space $C(X)$, we need to derive the perpendicular projection operator P_X. Where our model is of the form $Y = X\beta + e$. $\hat{Y} = P_X Y$ such that $\hat{Y}$ is our projection. The least squares solution to our model is essentially minimizing the error $||e||^2 = ||Y - X\beta||^2$ i.e the euclidean norm.
   By definition $X^T e = 0$ since the residuals are orthogonal to the design matrix. See Equation 2, our result is:
   $$P_X = X(X^T X)^{-1} X^T$$
   We can verify that P_X satisfies the key properties of a perpendicular projection operator:
   - Idempotent: $P_X P_X = X(X^T X)^{-1} X^T X(X^T X)^{-1} X^T = X(X^T X)^{-1} X^T = P_X$
   - Symmetric: $P_X^T = (X(X^T X)^{-1} X^T)^T = X(X^T X)^{-1} X^T = P_X$

   Using our design matrix X, we compute P_X and find that trace$(P_X) = 2$, which equals the dimension of the column space $C(X)$. This confirms the property that for any projection matrix, trace$(P_X) = \dim(C(X))$. The trace represents the sum of the eigenvalues of the projection matrix, which equals the number of dimensions in the space onto which we're projecting.

   iii. Take the dot product, $\hat{Y} = P_X Y$.
   $\hat{Y}$ is the best possible approximation of Y within the column space $C(X)$. Out of all possible linear combinations of the columns of X, $\hat{Y}$ is the one that is closest to the actual observed Y. The column space of X contains every possible estimated value of any combination parameters. As such, it by definition also contains the least squares solution.

   iv. R_X is simply the identity minus the original PPO P_X. It is expectedly also idempotent and symmetric.

   v. To compute $\hat{e}$:
   ```
   e_hat = Rx @ Y
   dim_C_perpendicular = 40 - X.shape[1]
   = 38
   ```
   Our error space has 38 dimensions since 2 are parameters.

   vi. The angle between our error and predictions is 90 degrees. This is expected since the residuals are the squared distances between our true values and predicted.

   vii. Where β is the vector of true unknown parameters (true means), and $\hat{\beta}$ is best estimated means, we can derive the general formula for $\hat{\beta}$ similar to 1.C ii. We begin with the orthogonality definition $X^T e = 0$ and our result is $\hat{\beta} = (X^T X)^{-1} X^T Y$ (See Equation 3).
   This is called the least squares estimate because it minimizes the sum of squared errors. Where the squared euclidean norm of e is $\|e\|^2 = e^T e = (Y - X\hat{\beta})^T (Y - X\hat{\beta})$.

```
Estimated Beta_1: 1.49, Estimated Beta_2  2.03
```

viii. Where $\hat{e}$ is the estimated residuals, not to be confused with e (the true error, i.e noise), we can analyze our residuals to get an idea of how much inherent variability exists in our data. Taking the square of the errors and dividing by degrees of freedom gives us the average of the squared errors (MSE), $\sigma^2 = 0.0365$, $\sqrt{\sigma^2} = 0.191$ which is close to our true standard deviation of 0.2.

ix. Our estimated coefficient standard deviations are expectedly the same. $\hat{\beta_{1\sigma}} = 0.0527, \hat{\beta_{2\sigma}} = 0.0527$

x. Our null hypothesis is that differences between the means of our groups is 0, $H_0 : \beta_1 - \beta_2 = 0$. To express this as $\lambda^T \beta = 0$, set $\lambda_1\beta_1 + \lambda_2\beta_2 = 0$. As such, λ must be $[1, -1]^T$. The reduced model is now $Y = X_0\beta_0 + e$.

xi. We can estimate the F-statistic by computing the reduced and full SSRs (essentially ANOVA). Our reduced model naturally has an additional DoF (39), since we have removed a parameter.

```
F-statistic: 53.550616027363525
```

We can reject the null hypothesis based on this high statistic. Our distribution is F(1,38).

xii. To see which group has a higher mean, we compute the t-statistic: -7.317, which is indeed the same as part b (again, the first group's mean is lower than the second).

xiii. Our model parameters are, X a matrix indicating group assignment, and $\hat{\beta}$ a vector of coefficients which is the best estimate of the means achieved by minimizing the squared error between Y and our predictions $\hat{Y}$. β, the true means, (ground truth values) should be $\beta_1 = 1.5$, $\beta_2 = 2.0$.

xiv. If we plug in our true means as coefficients, the SSR is 0.0228, which is naturally very low. If we were to increase the number of samples, this error would start to approach 0.

xv. Plugging in e (calculated in xiv. using true means), and computing the dot product: $\hat{e} = R_X Y$ the error between e and $\hat{e}$ is 2.6×10^{-30}, basically zero.

(d) Model with intercept: $Y = X_0B_0 + X_1B_1 + X_2B_2 + e$

i. Our design matrix (Equation 4) now consists of an additional column of ones to account for the intercept. $dim(X) = 2$, we maintain 2 linearly independent columns.

ii. Using **numpy.linalg.pinv** we can compute the Moore-Penrose inverse to solve for P_X, which is again a 40x40 matrix. It is essentially the same as the one in Part 1.c ii. (albeit with some long floats near zero, 3.469e-18, probably a result of the pseudoinverse).

iii. Our $H_0 : B_1 - B_2 = 0$ remains the same. Now $\lambda = [0, 1, -1]^T$

iv. Our t-statistic is again the same, -7.138.

v. In our intercept model, B_0 is the reference mean, and B_1 and B_2 are now the effect of being in their respective groups relative to the reference.

(e) Model without group 2 membership: $Y = X_0B_0 + X_1B_1 + e$

i. See the design matrix (Equation 5), $dim(X) = 2$. That is, 2 linearly independent columns.

ii. For this model, we test the hypothesis that $B_1 = 0$, so our contrast vector is $\lambda = [0, 1]^T$. This tests whether group 1 differs from the baseline represented by B_0.

iii. Our t-statistic is -7.138, which remains consistent with previous models. This demonstrates the equivalence of different GLM formulations for testing the same hypothesis.

iv. In this model, B_0 represents the mean of group 2 (since we removed X_2), and B_1 represents the difference between group 1 and group 2. This parameterization directly encodes the comparison we're interested in.

(f) Model representing overall mean: $Y = X_0B_0 + e$

This model has a design matrix consisting of only a column of ones, with $dim(X) = 1$. (see Equation 6) With this model, we cannot test the hypothesis of different group means, as we've removed all group-specific parameters. The model assumes all observations come from the same population with mean B_0. B_0 represents the overall mean across all observations. The simplest model is too constrained to detect group differences.

2. Paired t-test for comparing means from repeated measurements:

(a) Treat sample data as coming from same group of subjects.

i. Using SciPy's `stats.ttest.rel` which tests the H_0 that repeated samples have identical expected values, our t-statistic and p value have increased $t = -9.03, p = 2.65 \times 10^{-8}$. Our p-value still tells us we have different means, they are just different ways of doing the same thing.

(b) Compute the statistic using the GLM model:

i. The design matrix (Equation 7) has a $dim(X) = 21$. A column of ones (intercept), a column of time points, and n columns for subject ownership. We have many more linearly independent columns than before.

ii. Deriving the contrast vector is slightly more complex (see Eq. 8) to account for the difference between subjects. our contrast vector is $\lambda = [0, 1, 0, 0, \ldots, 0]^T$.

iii. We get the exact same $t = 9.029595$, though it is positive (negligible, magnitude is all that matters). We are indeed testing the same hypothesis.

Part 2: Permutation Testing

1. Single permutation test:

   (a) The data is generated the same way but with smaller sample sizes (see Fig. 2). Using `stats.ttest_ind` we get $t = -4.468, p = 7.67e \times 10^{-4}$.

   (b) Compute the permutation-based p-value:

      i. `D = np.concatenate([samples_1, samples_2])`

      ii. Using `itertools.combinations` we can replicate `nchoosek` functionality. Such that we generate 3003 valid permutations.

      iii. We can compute t-statistics for each permutation and plot the distribution (see Fig. 3) and compute the ECDF using `stats.ecdf` of our t-statistics.

      iv. Using a two-sided test to compute the empirical p value, we count the number of t-statistics that are more extreme than our original. `(np.abs(D_t_stats) >= np.abs(original_t_stat)).sum()`. We take the ratio of extreme counts (2) / total counts (3003), our $p_{emp} = 6.66 \times 10^{-4}$ Our original t-test assumes that we have a t-distribution, whereas the empirical permutation calculation builds the null distribution directly from our data. Such that we have a more accurate p value (i.e lower), which makes sense because our means really *are* different.

   (c) Repeating the process but using the raw mean difference as test statistic, we get $\mu_{diff} = -0.544$, our permutations lead to only 2 extreme mean differences, and our $p_{emp} = 6.66 \times 10^{-4}$ (see Fig. 4). This reinforces our findings further, but simply with a different t-statistic. We assume this repetition is just to show how permutation testing allows for more flexibility, but is computationally "expensive".

   (d) Compute approximate permutation-based p-value using 1000 random permutations.

      i. We can compute 1000 t-statistics (including the original) with `np.random.permutation(n)` where `n` is the total count of samples (14), and then split them into two groups maintaining sizes 6 and 8. For each permutation we compute the same t-statistic `stats.ttest_ind` and calculate the two-sided p-value by extracting extreme t-statistics. $p_{approx} = 2.00 \times 10^{-3}$ Given that we had 2 extremes out of 1000 permutations. (Fig. 5)

      ii. $p_{approx} = 2.00 \times 10^{-3}$ $p_{emp} = 6.66 \times 10^{-4}$ $p_{ttest} = 7.67e \times 10^{-4}$. Our p-value has naturally increased (with regards to the other two), since we have fewer permutations to draw from our, probability of rejecting the H_0 has decreased.

      iii. 14.8% of our permutations are replicas. The effective number of permutations is less than the nominal 1000. The error in p-value estimation due to duplicates is roughly proportional to the duplication rate.

2. Multiple comparisons correction:

   (a) We can load our data and plot the masks (see Fig. 6) and the masked FA maps (see Fig. 7). With our GLM for comparing means $Y = X_1\beta_1 + X_2\beta_2 + e$ we compute the t-statistic for each voxel (see Fig. 8). Our maximum in the ROI is $t_{max} = 6.5294$.

   (b) For 2 groups of 8 subjects each, we have 12,870 permutations for the group labels. For calculating the max statistic per voxel per permutation, I made use of `joblib` to parallelize computations. Additionally, I reduced the floating point precision from 64 to 16. Given that we are just making statistical comparisons, raw precision is not necessary. The ECDF and distribution of max statistics can be seen in Fig. 9.

   (c) Using the same method as in b, our $p_{perm} = 0.0920$, this is lower than our cutoff of 5%, so we are unable to reject the H_0, the means are not significantly different among groups.

   (d) The maximum t-statistic threshold given a p-value of 5% is 6.9396. This is done by computing the 95th percentile of the empirical distribution of maximum t-statistics. This was calculated by sorting all maximum t-statistics from the permutation tests and finding the value at the 95% position in the sorted array (specifically, at index $\lfloor 0.95 \times n \rfloor$ where n is the number of permutations)

1 Appendix

1.1 Equations

1. GLM Design Matrix

$$X = \begin{bmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 0 & 1 \\ \vdots & \vdots \\ 0 & 1 \end{bmatrix}$$

2. Solving for the PPO:

$$\begin{aligned} \overbrace{X^T e}^{\text{orthogonal definition}} &= 0 \\ X^T \overbrace{(Y - X\hat{\beta})}^{\text{definition of } e} &= 0 \\ X^T Y - X^T X \hat{\beta} &= 0 \\ X^T X \hat{\beta} &= X^T Y \\ \overbrace{\hat{\beta}}^{\text{solve for beta}} &= (X^T X)^{-1} X^T Y \\ \hat{Y} &= X\hat{\beta} = X(X^T X)^{-1} X^T Y \\ P_X &= X(X^T X)^{-1} X^T \end{aligned}$$

P_X is the transformation applied to Y

3. Derive the general formula for $\hat{\beta}$:

$$X^T e = 0 \tag{1}$$
$$X^T (Y - X\hat{\beta}) = 0 \tag{2}$$
$$X^T Y - X^T X \hat{\beta} = 0 \tag{3}$$
$$X^T X \hat{\beta} = X^T Y \tag{4}$$
$$\hat{\beta} = (X^T X)^{-1} X^T Y \tag{5}$$

4. Design matrix for model $Y = X_0 B_0 + X_1 B_1 + X_2 B_2 + e$

$$X = \begin{bmatrix} 1 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 1 \end{bmatrix}$$

5. Design matrix for model $Y = X_0 B_0 + X_1 B_1 + e$

$$X = \begin{bmatrix} 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \\ 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \end{bmatrix}$$

6. Design matrix for model $Y = X_0 B_0 + e$

$$X = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

7. Design matrix for model $Y = X_0\beta_0 + X_1\beta_1 + \sum_{i=1}^{N} S_i s_i + e$,

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & 1 \\ 1 & 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

8. Constrast vector for paired GLM:

$$\begin{aligned} 20\hat{\beta}_0 + \sum_{i=1}^{20} \hat{s}_i - (20\hat{\beta}_0 + 20\hat{\beta}_1 + \sum_{i=1}^{20} \hat{s}_i) &= 0 \\ -20\hat{\beta}_1 &= 0 \\ \hat{\beta}_1 &= 0 \\ \lambda &= [0, 1, 0, \ldots, 0]^T \end{aligned}$$

1.2 Figures

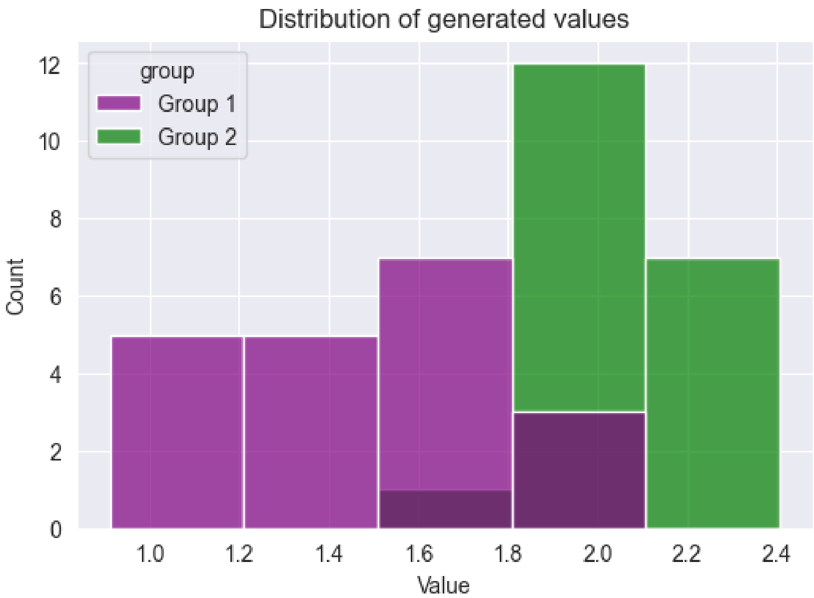

Figure 1:

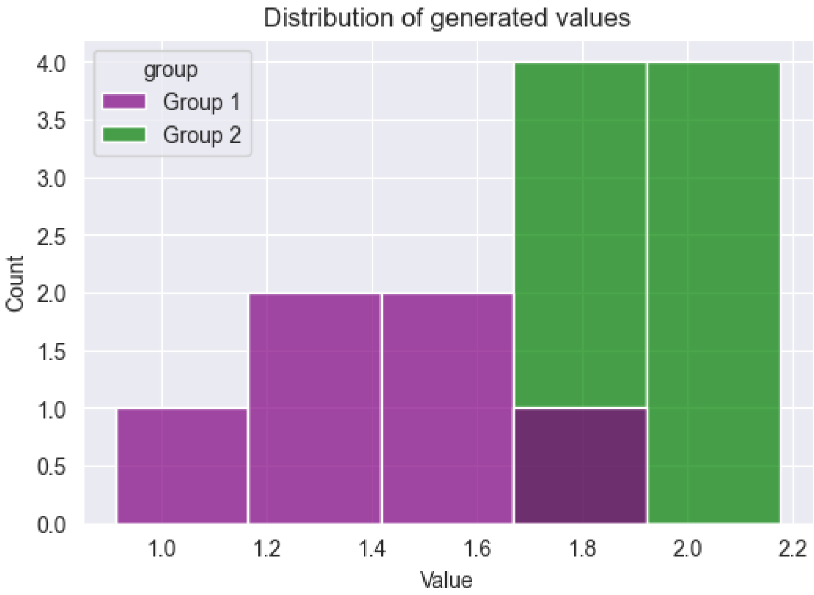

Figure 2: Reduced samples sizes leads to a less normal distribution

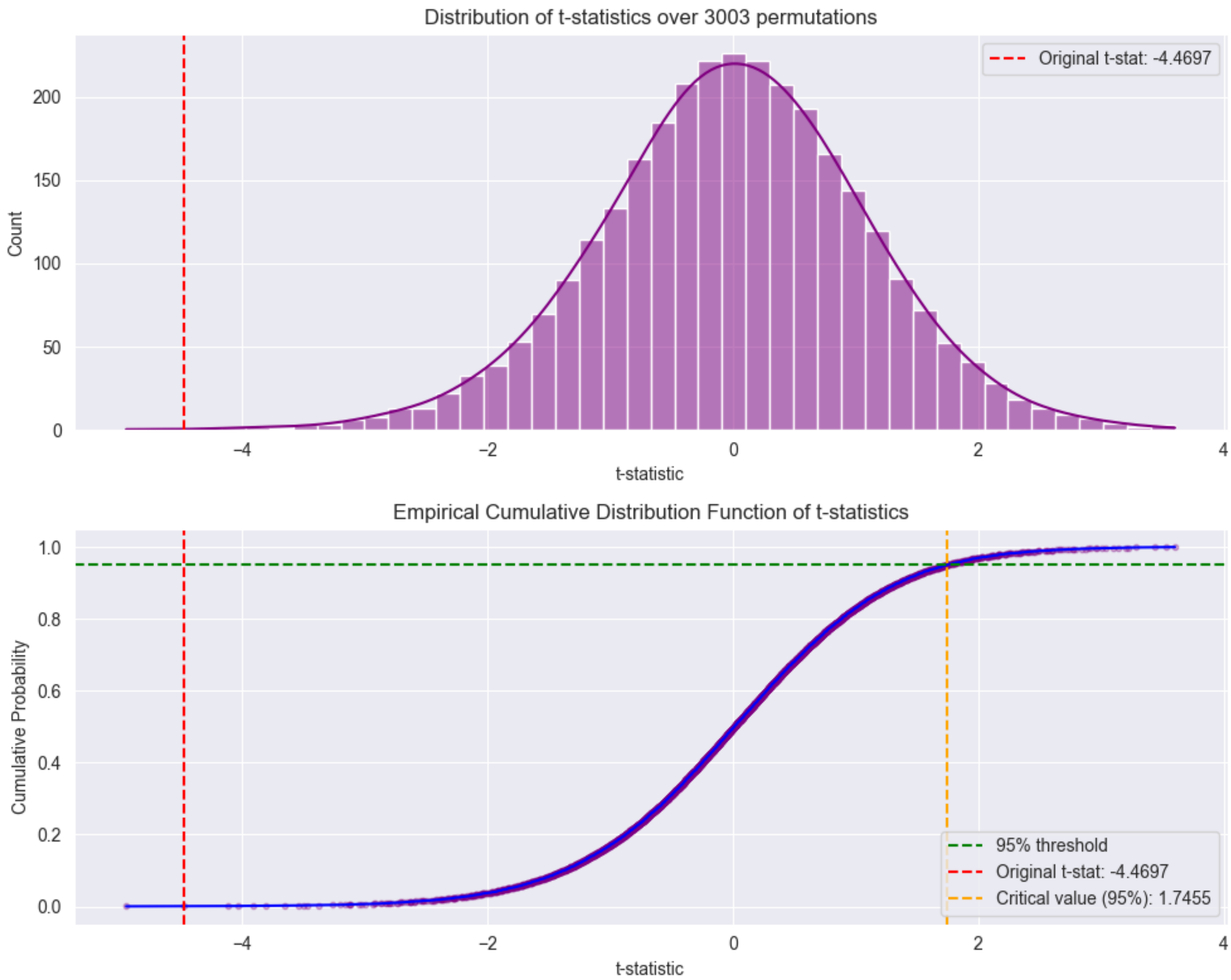

Figure 3:

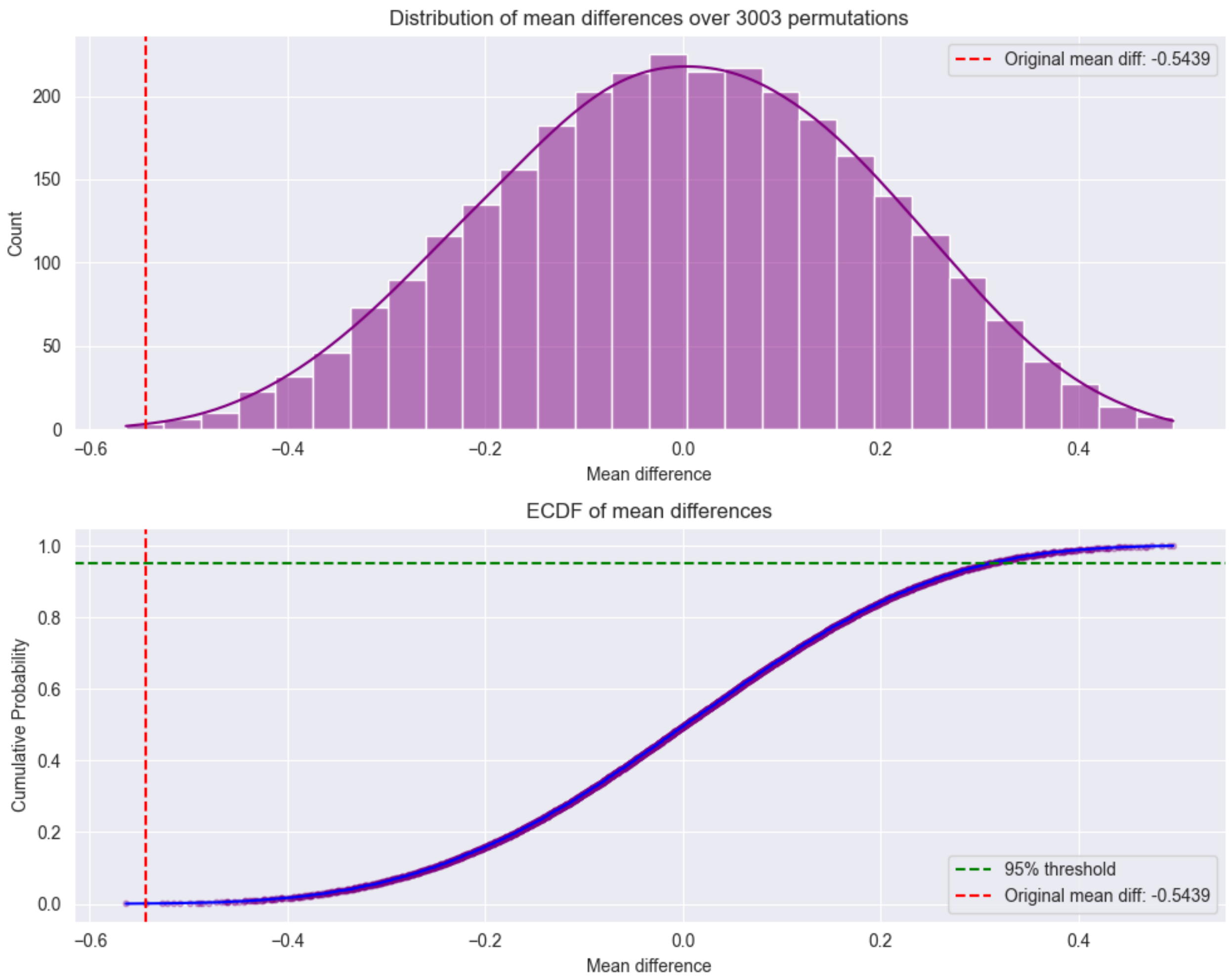

Figure 4:

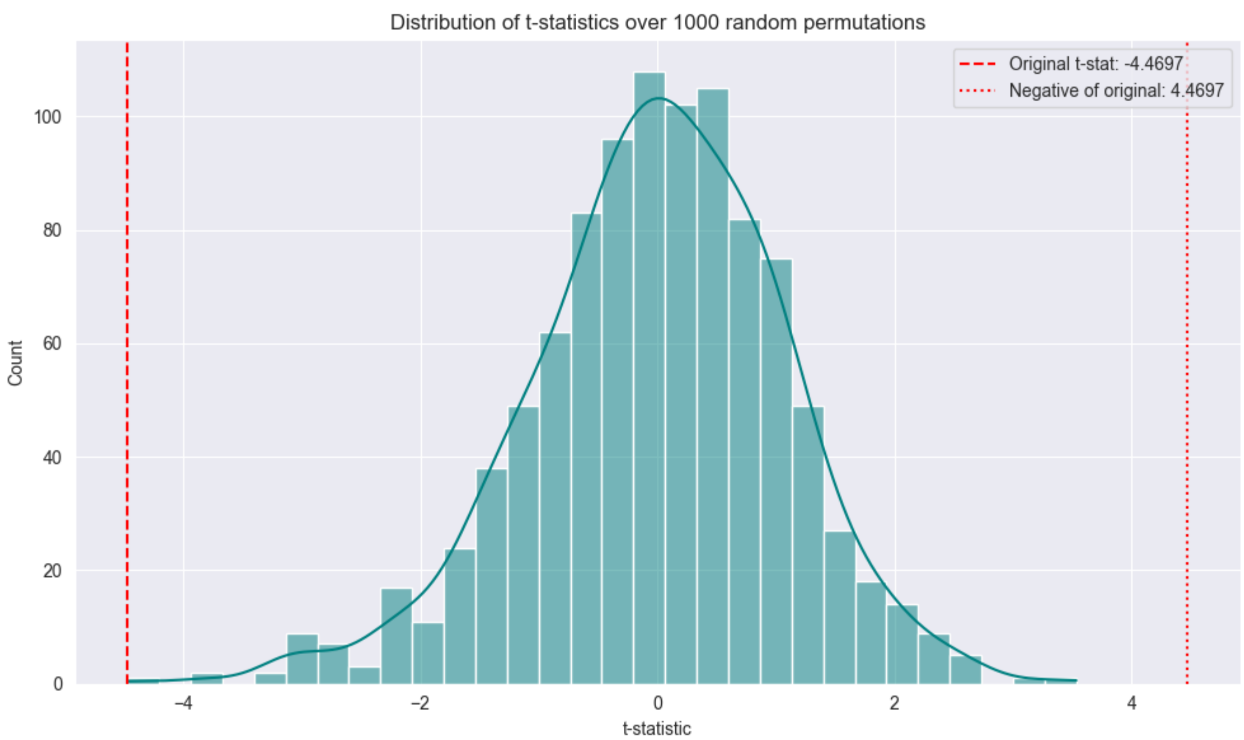

Figure 5:

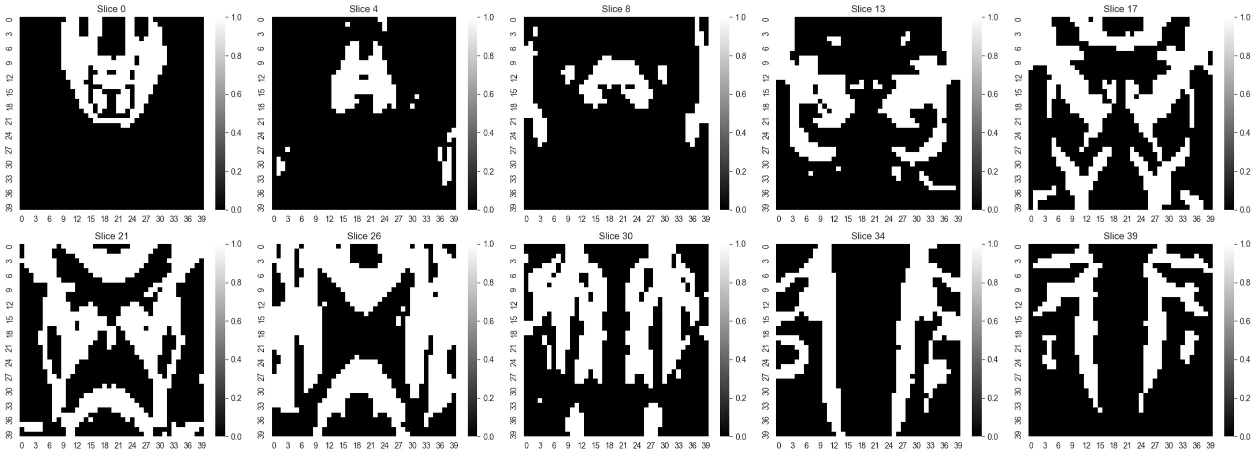

Figure 6: Mask maps for different slices in subject CPA 0

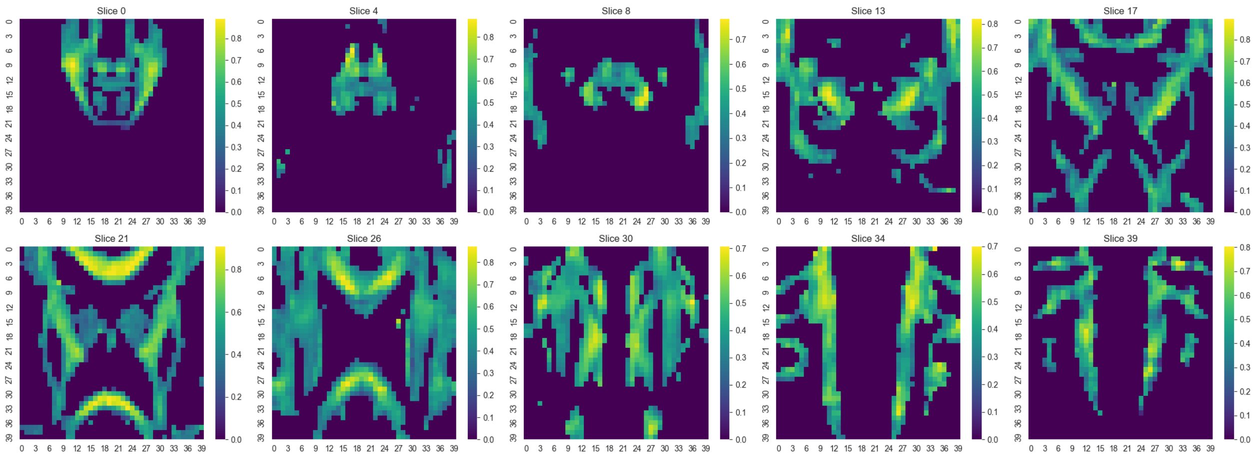

Figure 7: Masked FA maps for different slices in subject CPA 0

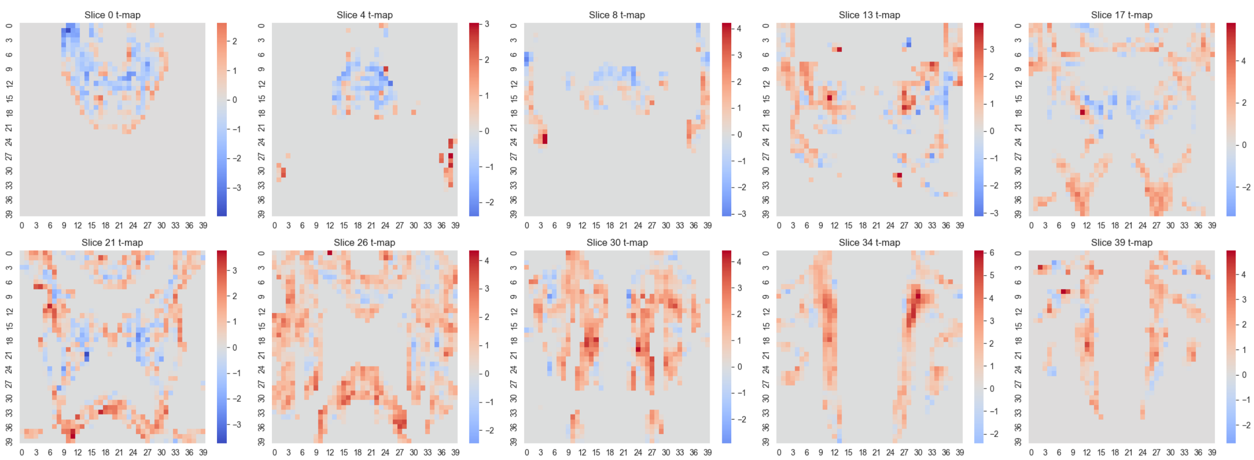

Figure 8: Per voxel t-statistic plots for a number of different slices in subject CPA 0

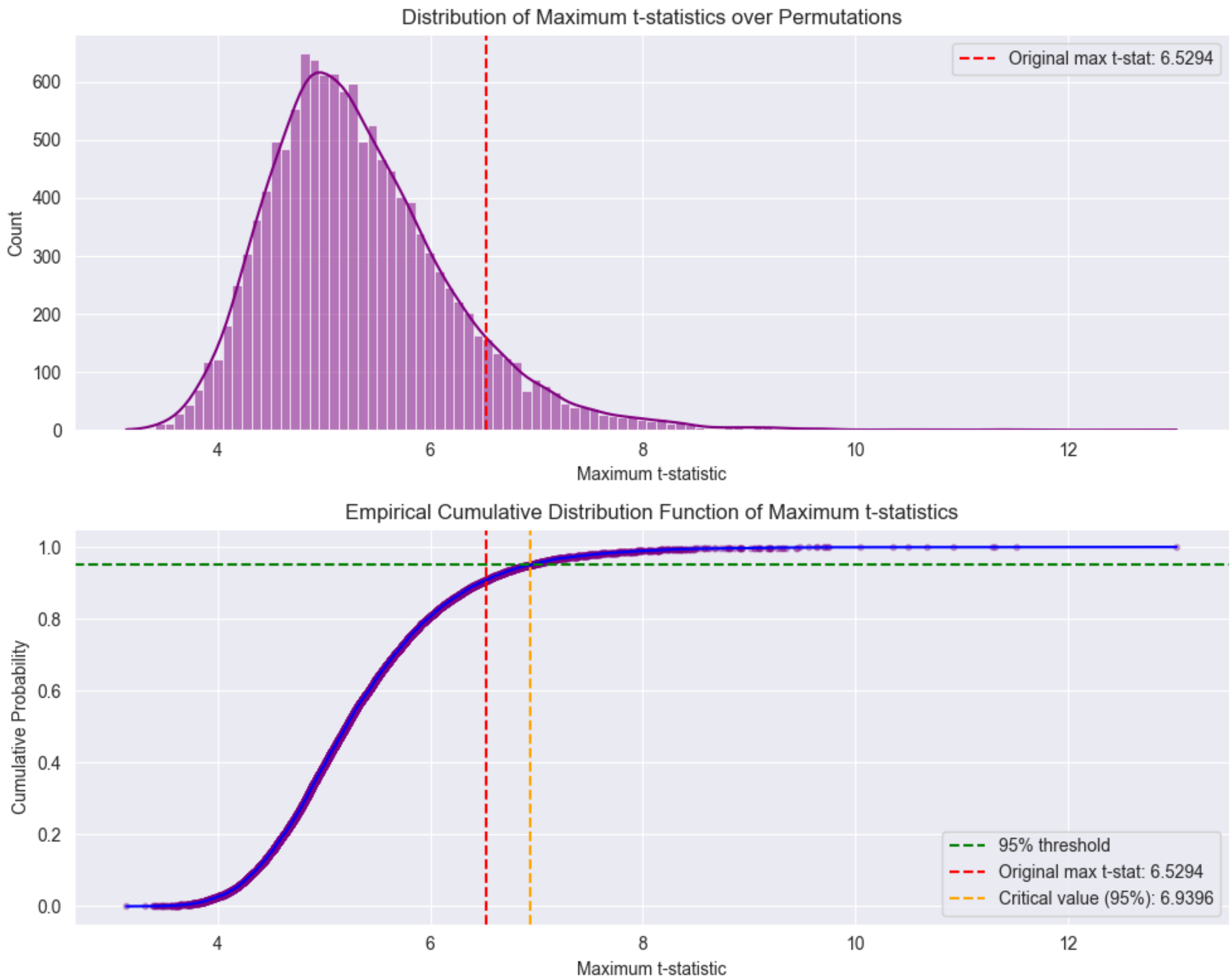
Distribution of Maximum t-statistics over Permutations
Original max t-stat: 6.5294
Count
Maximum t-statistic
Empirical Cumulative Distribution Function of Maximum t-statistics
Cumulative Probability
95% threshold
Original max t-stat: 6.5294
Critical value (95%): 6.9396
Maximum t-statistic

Figure 9: